HARSHITHA PUCHALAPALLI

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PROFESSIONAL SUMMARY

Software developer with hands-on experience across backend, full-stack, and applied-AI systems. Delivered a SQL Server execution-plan analyzer that reduced query cost by ~87% and a retrieval-augmented pipeline that grounds LLM output in retrieved source text. Skilled in Python, FastAPI, Flask, SQL, REST APIs, Docker, and pytest.

EDUCATION

Sri Venkateswara College of Engineering 2022 – 2026
B.Tech in Electronics and Communication Engineering **CGPA: 9.3/10**
Narayana Junior College 2020 – 2022
Class XI & XII **CGPA: 9.33/10**

PROJECTS

Clause Compass — Rental Agreement Risk Analyzer | Python, FastAPI, React, FAISS, Gemini API | Live Demo

- Built a FastAPI + React tool that ingests rental agreement PDFs, splits them into clauses, and extracts rent, deposit, lock-in, and notice terms.
- Rule engine flags risky clauses; LLM explanations are grounded in tenancy-law text retrieved via embeddings and FAISS, with citation checks and cached calls.
- Added personal-data redaction before any LLM call, a leave-early cost simulator, and a PDF viewer with coordinate-based clause highlighting.

SQL Query Doctor — Execution Plan Analyzer | Python, Flask, SQL Server | Live Demo

- Built a Flask tool that turns SQL Server execution plan XML into an interactive operator graph with per-operator cost shares.
- Rule engine flags table scans, key lookups, missing indexes, plan warnings, and bad row estimates; on the demo slow query its recommended index is estimated by SQL Server to cut query cost and runtime by ~87%.
- Gemini API generates plain-English fixes with caching and model fallback, backed by upload validation and a one-click sample plan.

Tirumula Dairy Management System | Supabase, HTML, CSS, JavaScript | Live Demo

- Built a full-stack dairy management app (Supabase PostgreSQL, HTML/CSS/JS) with an admin dashboard and public product catalog.
- Added authentication with role-based access, plus CRUD, image upload, search, filtering, and category management for inventory.
- Implemented a POS billing system that updates stock automatically and generates PDF invoices, with sales analytics.

NutriGaze: Fresh vs Rotten Fruit Classification | Python, Flask, TensorFlow, OpenCV

- Built a TensorFlow + OpenCV deep-learning app that classifies fresh vs rotten fruit with 97% accuracy.
- Designed a responsive Flask interface for real-time image upload, showing predictions with confidence scores.
- Wired the trained model into the backend for fast end-to-end inference.

SKILLS

GenAI & ML: RAG, LLM Integration, Gemini API, TensorFlow, OpenCV
Languages: Python, SQL, JavaScript
Backend & APIs: Flask, FastAPI, REST APIs, SQL Server, PostgreSQL, Supabase
DevOps: Docker, CI/CD, GitHub, pytest
Frontend: React, HTML, CSS
Tools: Power BI, Excel, MATLAB

CERTIFICATIONS

- Generative AI Professional — Oracle
- Artificial Intelligence — IBM